Response: Dabrowski Hilbert–Smith preprint

Sherlock reverse-verification notes

Alan J. Coppola (AI-assisted / AI-generated using Cursor, with Grok, Anthropic, and OpenAI models)

2026-10-08

Independent read of ***Integral tail signatures and the Hilbert--Smith conjecture***
Upstream: `github.com/adbrw/HS_proof` SHA: `abb0c44…`
PDF SHA-256: `e305fd47…` Repo: `github.com/ajjcoppola/hilberts-5th-oai`

This work is AI-assisted and AI-generated using Cursor, with Grok, Anthropic, and OpenAI models, under the direction of Alan J. Coppola.

Original questions

1. Overview: how faithful p-adic actions are excluded near the identity homeomorphism.
2. Geometric argument, or algebraic invariants in every Euclidean neighborhood?
3. Public / internet responses to this AI-generated proof?
4. Categorize the proof; what does it say about the structure of $\mathbb{R}^n$?
5. Hold/break under Menger-type sponges as $\mathbb{R}^n$ replacements (where the theorem is false)?
6. Is the contradiction that $\mathbb{R}^n$ cannot factor through an orbit space with "wonky" local sphere maps?

TL;DR

Dabrowski's preprint claims a proof of the Hilbert–Smith conjecture by ruling out effective continuous $\mathbb{Z}_p$-actions on connected finite-dimensional manifolds (boundary allowed). The argument is not Yang's orbit-dimension obstruction and is not Pardon's three-manifold mapping-class argument. It is an integrality-versus-p-divisibility contradiction for an *integer signature tail* in ultimate quadratic L_4: a small open tail of $\mathbb{Z}_p$ Haar-averages a degree-one pinch in a Euclidean chart; finite quotients and a character detector produce a duality-compatible homotopy idempotent on $S^n \times \mathbb{CP}^2$; iterated Karoubi localization boundaries recover one copy of $\mathbb{CP}^2$, giving a class whose ordinary signatures are eventually 1. Independently, on a fixed compact free C_p-space, permutation-tensoring is locally p copies of the identity and is nilpotent after Mayer–Vietoris, so every L_4 class has eventually p-divisible signatures. Hence $1 \in p\mathbb{Z}$. This is the same obstruction family as the OpenAI (2026-09-23) Witt-sheaf argument, with different packaging. A targeted audit found no outright failure in the load-bearing PDF lemmas; several steps remain weak (negative K / tail extraction, nilpotence of ν, the graph corner, $\mathbb{CP}^2$ evaluation, and an independently printed Lean axiom list). On Menger-type sponges, where faithful $\mathbb{Z}_p$ actions exist, the writeup correctly refuses to run because chart

and combinatorial dual-cell hypotheses fail. The claimed Lean 4 certificate was not rerun in this session.

Proof overview (Q1, Q2, Q4, Q6)

Proof overview — Dabrowski Hilbert–Smith preprint (2026-10-08)

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Answers to reverse-verification questions Q1, Q2, Q4, Q6, applied to A. Dabrowski's *Integral tail signatures and the Hilbert–Smith conjecture*. Source: vendored PDF in `paper/HSproof.pdf`.

Statement

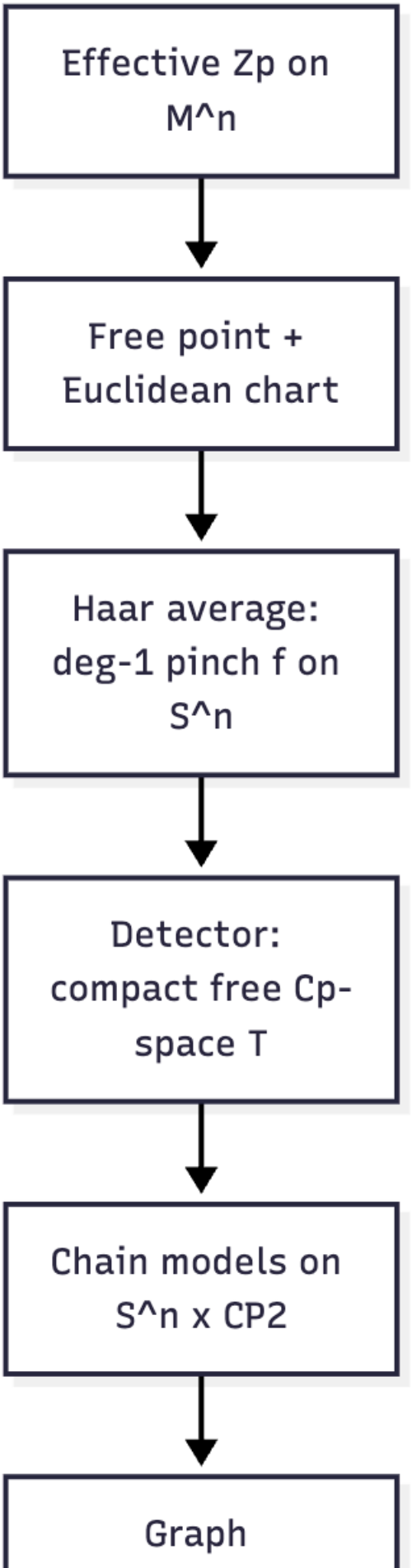
Theorem 1.1 (p-adic exclusion). For every prime p, a continuous action of $\mathbb{Z}_p$ on a connected finite-dimensional manifold has nontrivial kernel.

Corollary 1.2 (Hilbert–Smith). A second-countable locally compact Hausdorff group acting continuously and effectively on such a manifold is a Lie group with its given topology.

Manifolds are Hausdorff and second countable, and **may have boundary**. All actions are jointly continuous. Corollary 1.2 is the standard reduction: locally compact structure + Newman ([Pa19]) + Theorem 1.1 + Gleason–Yamabe (no small subgroups ⇒ Lie). The Lean `Challenge.lean` states the **boundaryless** form; the paper recovers the boundary case by restricting to the interior (homeomorphisms preserve the interior by local homology).

Mechanism in one paragraph

Assume an effective $\mathbb{Z}_p$-action on a connected n-manifold, $n \geq 1$. A free point exists (Lemma 8.1). Center a Euclidean chart; Haar-average coordinates of a small open tail $H = p^j\mathbb{Z}_p$ to get an invariant degree-one pinch $f : S^n \to S^n$. A character detector embeds a compact free C_p-space T (odd sphere) from the free covering Z^+/H'. On the product Poincaré manifold $W = S^n \times \mathbb{CP}^2$, shrinking triangulations supply controlled chain models; finite quotients $G_i = H/K_i$ act up to homotopy. Compression to an exterior band, then a graph idempotent E, splits in Karoubi completion to a Poincaré corner $d \in L_{n+4}(B_{G,Z}(T \times S^n))$. Forgetting equivariance recovers the scalar germ of f. Iterated Karoubi localization boundaries Δ_T cut the sphere down to a point and retain one copy of $\mathbb{CP}^2$, giving $\alpha \in L_4(A_G(T))$ with **signature tail constantly** 1 (Proposition 13.1). Independently, on any compact free C_p-space T, tensoring with the positive permutation form $\tau_i = \mathbb{Q}[G_i/P_i]$ is locally p copies of the identity; controlled Mayer–Vietoris makes $\nu = p \cdot \mathrm{id} - \tau \otimes (-)$ **nilpotent**. Equivariant signature characters plus cyclotomic arithmetic force every class in $L_4(A_G(T))$ to have **eventually p-divisible ordinary signatures** (Theorem 4.3). Hence $1 \in p\mathbb{Z}$.


Effective Zp on M^n
Free point + Euclidean chart
Haar average: deg-1 pinch f on S^n
Detector: compact free Cp-space T
Chain models on S^n x CP2
Graph

Q1 — Why no p-adic homeomorphisms near id?

Same geometric reason as OpenAI, packaged differently.

1. **Small open tails.** Effectivity plus Roberts's theorem on pointwise-finite families produce a free point (Lemma 8.1). Joint continuity then supplies $H = p^j\mathbb{Z}_p$ with arbitrarily small Euclidean displacements on a chart ball. Finer kernels $K_i = p^{\ell_i}\mathbb{Z}_p$, $\ell_i \to \infty$, give finite cyclic groups $G_i = H/K_i$ of unbounded p-power order.
2. **Averaging near id.** The Haar coordinate average $F_0(x) = \int_H hx\, d\mu(h)$ is H-invariant and C^0-close to the identity on a chart, so the compactified pinch f has degree one. Finite groups have no smaller open subgroup to average over while keeping degree one.
3. **Arithmetic uses the sequence, not a single cover.** Divisibility is a statement about the **tail** of $L_4(A_G(T))$: signatures of the coordinate forms (V_i, b_i) are eventually divisible by p. Realization produces the constant tail 1. That needs infinitely many finer G_i, which only $\mathbb{Z}_p$ supplies.

So the disproof is: a faithful action near the identity would manufacture a Poincaré class over a fixed free C_p-space whose ordinary signatures stay 1, while permutation-tensor nilpotence forces those signatures into $p\mathbb{Z}$.

Q2 — Geometric or algebraic?

Mostly algebraic L-theory, with a thin Euclidean shell.

Layer	Nature	Content
Chart / free point / Haar pinch	Geometric	Local Euclidean coordinates, Newman in the corollary, orientation of $S^n \times \mathbb{CP}^2$
Detector / sheets / triangulations	Topological / PL	Equivariant map $Z^+ \to T = S^{2m-1}$; mesh $\eta_i \to 0$ chain approximations
Duality / corners / localization	Algebraic	Ultimate lower quadratic L-theory $L_k = \pi_k L^{\langle -\infty \rangle}$; Karoubi filtrations [CP95]; Ranicki Poincaré pairs [Ran80I, Ran80II]; Balmer–Schlichting splitting [BS01]
Contradiction	Arithmetic	Integer signature tail: $1 \in p\mathbb{Z}$

The obstruction is **not** Yang/Bredon–Raymond–Williams cohomological-dimension raising. Orbit-space dimension never appears. The invariant that must hold in every Euclidean neighborhood is an **integer signature** realized by $\mathbb{CP}^2$ after forgetting a controlled equivariant corner.

Q4 — What the proof says about the structure of $\mathbb{R}^n$

Open sets in $\mathbb{R}^n$ (after compactification to S^n and product with $\mathbb{CP}^2$) carry:

1. **Controlled Poincaré–Lefschetz duality** at shrinking mesh, so algebraic Poincaré pairs exist with propagation $\to 0$.

2. **A degree-one invariant pinch** after Haar averaging of a small open tail, so the scalar L-class is the manifold germ of $S^n \times \mathbb{CP}^2$, not a multiple of p.
3. **No p-divisible integer signature tail** for that germ after transferring across a free C_p-space T built from local sheets.

In slogan form: $\mathbb{R}^n$ is **signature-tail rigid** under continuous p-adic orbit quotients. The rigidity is measured in $L_4(A_G(T))$, not on the possibly wild orbit space.

Q6 — Is the contradiction "orbit space factoring + wonky local sphere maps"?

Partly yes, but the wonky maps are controlled chain maps, not sphere maps of unbounded degree.

- Yes: the degree-one pinch f is H-invariant, so it factors through the orbit space of the tail. Finite quotients G_i act on chain models up to homotopy (graph type).
- The "wonky" part is compression to an exterior band (Lemma 10.1) and a duality-compatible homotopy idempotent E whose Karoubi corner forgets to m_f.
- The **basic contradiction** is arithmetic on integer signatures:

$$\sigma_{\text{tail}}(\alpha) = [(1,1,1,...)] \qquad \text{and} \qquad \sigma_{\text{tail}}(\alpha) \in p\mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})},$$

  hence $1 \in p\mathbb{Z}$.

Orbit-space factorization is necessary but not sufficient. The paradox is that factorization plus faithfulness (unbounded p-power finite quotients) plus Poincaré corners would force an integral class of signature 1 to be eventually p-divisible.

Section map

Sections	Role
1	Statement; two-part outline; conventions (rational chains, integral L-groups)
2	Asymptotic categories $A_G(X)$; support Karoubi filtrations
3	Equivariant signatures; Lemma 3.1 (cyclotomic); tail extraction σ_{tail}
4	Divisibility: local triviality of τ, nilpotence of ν, Theorem 4.3
5	Signs; compatible idempotent splitting; scalar normalization
6	Localization boundaries; union of Poincaré pairs
7	Acyclic carriers; dual-block Poincaré–Lefschetz at mesh η
8–9	Geometric data: chart, Haar pinch, detector T, chain models on $S^n \times \mathbb{CP}^2$

Sections	Role
10–11	Compression, graph corner, forgetting to m_f
12	Iterated boundary evaluation: $\Delta_1 m_f = \varepsilon[\mathbb{CP}^2]$
13	Realization of signature 1; proofs of 1.1 and 1.2

Audit ledger (Targets A–C)

Targeted audit ledger (Sherlock targets A–C)

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Scope: line-by-line on load-bearing lemmas only (plan: targeted audit).
Grades: `PASS` = argument appears complete and correctly cited; `WEAK` = plausible but citation gap, omitted routine detail, hypothesis that needs expert/Lean check, or certificate not independently rerun; `FAIL` = identified break.
Overall targeted verdict: No FAIL found on the PDF's load-bearing lemmas. Several `WEAK` items remain — especially **Target C** (Lean certificate vs published theorem), which this audit did **not** compile. The proof does **not** "prove too much" on known sponge models (see `03_sponge_stress_test.md`).

Target A — Signature-tail lattice / integer signatures independent of p-power

The arithmetic engine is coarser than OpenAI's $L_d^{-1}\mathbb{Z}$ lattice: ordinary signatures of rational symmetric forms are **integers**, and the tail map lands in $\mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})}$. The "bounded denominator" claim is just integrality of σ, plus that two lifts of an $L_4(V_G)$ class agree eventually.

Lemma 3.1 — Cyclic arithmetic

PDF: §3, p. 3
Lean: `AlgebraicReduction.lean`, `EquivariantSignature.lean`
Grade: PASS

- If $G = C_{p^a} = \langle g \rangle$ and $\mathrm{Sign}_G(V,b)(g) = 0$, then $p \mid \sigma(V,b)$.
- Write $\mathrm{Sign}_G = \sum m_j \chi_j$; $F(t) = \sum m_j t^j$ is divisible in $\mathbb{Z}[t]$ by Φ_{p^a}, so $F(1) = \sigma$ is divisible by $\Phi_{p^a}(1) = p$.
- Standard cyclotomic fact; Lean proves the polynomial form (`prime_dvd_eval_one_of_primitive_root`) and the character form. No residual risk here.

Proposition 3.2 — Tail extraction

PDF: §3, pp. 3–4
Grade: WEAK (depends on negative K of $\mathbb{Q}[G]$)

- Semisimple rings $\mathbb{Q}[G_i]$ have vanishing negative K-groups [Wei13, III.4]; Bass's polynomial definition commutes with the finite-support subcategory S, so $K_{-k}(S) = 0$ for $k \geq 1$.
- Rothenberg comparison $\Rightarrow L_j(S) = \bigoplus_i L_j^p(\mathbb{Q}[G_i])$. Karoubi filtration by finite index sets lifts every $L_4(V_G)$ class to $L_4(P)$; two lifts differ by $L_4(S)$, so coordinate classes agree eventually.
- **Why WEAK:** this is exactly the `NegKHighAll` / decoration-map bijectivity package that occupies most of `HSFormal/LTheory/` (( )104 modules). The PDF cites Weibel as a black box for $K_{-k}(\mathbb{Q}[G]) = 0$. If that comparison fails for the *asymptotic sequence category* (not just a single group ring), the tail map is not well-defined as a homomorphism to the product-modulo-sum. Lean claims to prove the iterated bounded-category form; this audit did not check those modules line-by-line.

Theorem 4.3 — Integer tail divisibility

PDF: §4, p. 4
Grade: PASS (with WEAK dependence on Lemma 4.2 and Prop. 3.2)

- Extract forms (V_i, b_i). Nilpotence $\Rightarrow (p - \tau_i)^s [V_i, b_i] = 0$ in $L_4^p(\mathbb{Q}[G_i])$ for large i.
- Character of τ_i at a generator g_i vanishes, so $p^s \operatorname{Sign}_{G_i}(V_i, b_i)(g_i) = 0$, hence that character is zero (complex numbers), and Lemma 3.1 gives $p \mid \sigma$.
- The arithmetic implication is clean (`character_vanishes_of_prime_power_annihilation` in Lean). Load-bearing gaps sit in the geometric nilpotence and in tail extraction, not here.

Proposition 13.1 / Prop. 12.4 — Realization of signature 1

PDF: §§12–13, pp. 15–16
Grade: WEAK

- After n localization boundaries, one component of $T \times S^0$ retains exactly one copy of $\mathbb{CP}^2$, sign $\varepsilon \in \{1, -1\}$ independent of i. Middle form $\langle 1 \rangle$; half-unit quadratic lift has signature 1.
- Identity-character trace identifies ordinary signatures of equivariant forms with those of their scalar images, so $\sigma_{\text{tail}}(\alpha) = [(1, 1, ...)]$.
- **Why WEAK:** (i) sign bookkeeping through n cuts, one ordered diagonal, and component projection is the same class of foot-gun as OpenAI's cylinder signs; (ii) "point control + subdivision identifies the constant model" is sketched rather than expanded; (iii) the half-unit comparison $\varphi/2$ on rational Poincaré complexes is standard Ranicki but must survive every exterior quotient. No contradiction found on reading; this is the geometric–algebraic interface of realization.

Target B — Transfer / averaging multiplies by p without leaving the integral lattice

Here "integral lattice" means: $\tau \otimes (-)$ and $p \cdot \mathrm{id}$ act on the same ultimate L-groups of $A_G(T)$, and Haar averaging produces a **degree-one** (hence signature-one after $\mathbb{CP}^2$) class rather than a p-multiple.

Lemma 4.1 — Local triviality of τ

PDF: §4, p. 4
Grade: PASS

- On a compact $U \subset T/C_p$ inside a covering trivialization, a sheet function $c : L \to C_p$ gives an orthogonal equivariant isometry $\tau \otimes (-) \cong \mathrm{id}^{\oplus p}$ of propagation zero.
- Compact positively separated sheets $\Rightarrow$ controlled morphisms eventually preserve them. Standard covering-space linear algebra. Appears complete.

Lemma 4.2 — Nilpotence $\nu^s = 0$

PDF: §4, p. 4
Grade: WEAK

- Finite compact cover of T/C_p of size s, independent of i. Induct via support Mayer–Vietoris from Lemma 2.2: $\nu = 0$ on $A \cap B \Rightarrow \partial(\nu x) = 0 \Rightarrow \nu x$ comes from $L_*(A_A) \oplus L_*(A_B)$; induction kills the first summand by ν^{s-1}, local triviality kills the second.
- **Why WEAK:** the argument assumes the support filtration of Lemma 2.2 induces a genuine localization / Mayer–Vietoris sequence in **ultimate** L-theory, with ν acting naturally on the sequence. Carlsson–Pedersen [CP95, Thm 4.2] is cited later for boundaries; here the MV sequence is called "natural" from Lemma 2.2 without repeating the $L\langle -\infty \rangle$ hypotheses (functoriality, additivity, Karoubi localization with connecting maps — listed as conventions on p. 2). Specialists should check that ν commutes with ∂ strictly enough for the nilpotence induction, including at $s = 1$.

Lemma 2.2 — Support calculus

PDF: §2, pp. 2–3
Grade: PASS (with residual on "equality means eventual equality")

- Duality-invariant Karoubi filtration; $A_A \cap A_B = A_Y$; quotient equivalences; $A_G(S) \simeq A_S$; fully faithful when $d(S, Z) > 0$.
- Compactness of X is used to send labels approaching both A and B into Y. Relabeling by nearest-point projection has propagation $\to 0$.
- Residual: asymptotic categories identify morphisms only eventually. The filtration splits must be **uniform in the sequence index** for later nilpotence. The writeup addresses this via null sequences r_i and filtered orthogonal splits; looks carefully written.

Lemma 8.1 + Haar pinch (8.1)–(8.6)

PDF: §8, pp. 9–10
Grade: PASS

- Free point: nonzero closed subgroups of $\mathbb{Z}_p$ are the tails H_j; if all stabilizers were nonzero, discrete $\mathbb{Z}$ would have finite orbits, hence finite image by [Rob75], contradicting effectivity.
- Haar average F_0 is continuous, invariant, $|F_0(x) - x| \le 2D$; compactified pinch f has degree one by straight-line homotopy to the identity pinch s_ρ.
- Same geometric idea as OpenAI's `lem:averaged-test`, done in coordinates rather than as a map $S^d \to Y \to S^d$. Finite $\mathbb{Z}/p$ cannot produce a sequence K_i with displacements $\to 0$ while remaining a proper open subgroup of a finite group. **Sanity:** see sponge doc.

Graph idempotent / Lemma 5.1–5.2 / Prop. 10.2

PDF: §§5, 10–11
Grade: WEAK

- Homotopy action of G_i on (C, φ) produces $E = q_i^{-1} \sum_g A_g \otimes R_{g^{-1}}$ with $E^2 \simeq E$ and $E\Phi \simeq \Phi E^*$ in the **exterior quotient**.
- Balmer–Schlichting [BS01] splits E in $K^b(\operatorname{Kar} Q)$; normalized corner $\varphi_D = q^{-1} r \Phi r^*$. Scalar normalization (Lemma 5.2) says that after forgetting, the corner is isometric to (C, φ).
- Compression (Prop. 10.2) kills defects through an exterior band around radius t; detector cancellation $\chi(sg^{-1})\chi(g) = \chi(s)$ is the reason free sheets work.
- **Why WEAK:** (i) all homotopies, including products of $\sim q_i^2$ terms, are claimed to be uniformly controlled independent of rank — the paper emphasizes "uniform control rather than

a bound on the number of summands," which is the right slogan but easy to under-prove; (ii) splitting "once in the sequence category" must produce asymptotically small representatives; (iii) signs in (5.1) and the double dual identification are spelled out, but the graph-type calculus of §9–10 is the longest formal stretch. Lean's `CornerClass.lean` / `GraphCorner` is the matching formal target.

Target C — Lean certificate vs published theorem

Grade: WEAK (certificate not rerun; statement alignment looks good)

Claim (upstream README)	What this audit checked
`Challenge.lean` states classical HSC, proof `sorry`, Mathlib only	**PASS.** Boundaryless topological n-manifold; locally compact second-countable Hausdorff group; joint continuity; faithful action; conclusion C^∞ Lie in the **given** topology.
`Solution.lean` = `HSFormal.hilbertSmith n M G`	**PASS** as text: same type signature, one-line call.
`HSFormal.hilbertSmith` proved in `HilbertSmithNegK.lean` from `NegKHighAll`	**Text PASS, compile unchecked.** `negKHighAll := negKHighAll_of theoremB Reg.regKarStatement`, then `hilbertSmith_all.1`. Earlier comments in `HilbertSmithModel.lean` still describe `NegKHighAll` as "being formalized on the v4.35 port"; the terminal file `HilbertSmithNegK.lean` claims that induction is now a theorem.
Axioms used: `propext`, `Classical.choice`, `Quot.sound`	**UNVERIFIED.** This environment did not run `lake env lean` / `#print axioms`. A garbled third-party paste of the README listed `HilbertSmithNegK.lean` and even `HSproof.pdf` as axioms; the **vendored** README does **not**. Treat the three-axiom claim as a hypothesis until printed.
238 modules, import closure of `HilbertSmithNegK.lean`	**PASS** as a file count (`find HSFormal -name '*.lean' | wc -l` = 238). Full tree not vendored.
Comparator / SafeVerify "solution is okay"	**UNVERIFIED.** Scripts live upstream under `verify/`; not rerun (Linux Landlock, ( )15 GB).

Claim (upstream README)	What this audit checked
PDF ↔ Lean statement	**Mostly PASS.** Lean `Challenge` is boundaryless; paper Theorem 1.1 allows boundary and proves it by interior restriction. Lean also packages `HilbertSmithWithBoundary` and `PadicExclusionWithBoundary` in `Statement.lean`. Paper uses rational chain coefficients throughout; Lean builds a concrete `LowerLTheory` model. Residual: whether every PDF lemma is in the import closure, vs a weaker model with leftover `LowerLTheory` hypotheses. The **terminal** theorem `hilbertSmith` has no remaining hypothesis in the source text.

PDF vs Lean packaging. The PDF is 17 pages of controlled L-theory. The Lean tree is an engineering expansion: Newman, Gleason–Yamabe, Tao's Lemma 7, a colimit model of lower L-theory, and a high negative-K induction. That is expected. The risk is not "too many files"; it is that a `sorry` or extra axiom hides in `LTheory/Model/NegK/High/` after the README's axiom list was written.

Suggested axiom-print command (not run here):

```
lake env lean --stdin <<< 'import HSFormal.HilbertSmithNegK
#print axioms HSFormal.hilbertSmith'
```

Cross-cutting residual risks (not graded FAIL)

1. **Ultimate decoration** $L^{\langle -\infty \rangle}$ is used everywhere; projective-to-ultimate maps and idempotent-completion invariance are listed as required properties rather than proved in the PDF.
2. **Signs** in Poincaré–Lefschetz dual blocks (Lemma 7.4) and iterated boundaries (Prop. 12.4).
3. **Homology-manifold edge** is outside the theorem's hypotheses; see sponge doc.
4. **No independent `#print axioms`** in this session.

Suggested expert / Lean priorities

Priority	Object	Why
1	`#print axioms HSFormal.hilbertSmith` on a clean `lake build`	Distill C
2	Lemma 4.2 nilpotence vs [CP95] MV in ultimate L	Distill B

Priority	Object	Why
3	Prop. 3.2 tail extraction / `NegKHighAll`	Distill A/C
4	Prop. 10.2–11.6 graph corner + detector cancellation	Realization
5	Prop. 12.4 iterated boundary $\to \varepsilon[\mathbb{CP}^2]$	Signs / evaluation

Sanity check (finite $\mathbb{Z}/p$)

See `03_sponge_stress_test.md`. Failure is at the **unbounded tail** K_i with displacements $\to 0$: a finite group has no such sequence. Degree-one Haar averaging over a proper open subgroup also fails. Arithmetic of Theorem 4.3 is not reached. **Sanity PASS.**

Sponge stress test (Q5)

Sponge / non-manifold stress test (Dabrowski)

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Goal: Can the argument be run on spaces where faithful continuous $\mathbb{Z}_p$-actions **exist**, thereby "proving too much"? If every lemma's hypotheses hold on such a space, that is a candidate gap. If a manifold-specific hypothesis fails early, the proof is correctly scoped.

Framing: A Menger sponge where Hilbert–Smith is false does **not** refute a theorem about manifolds. It is a probe for overclaiming.

Sanity: finite $\mathbb{Z}/p$ on $\mathbb{R}^2$

Step	Status
Chart / free point	OK for a free rotation (or fails faithfulness differently)
Unbounded tail $K_i = p^{\ell_i}\mathbb{Z}_p$ with displacements $\to 0$	**FAILS** (group is finite)
Haar degree-one pinch through a proper open subgroup	**FAILS**
L_4 tail divisibility / $\mathbb{CP}^2$ realization	Not reached

Why the tail fails. Realization needs a sequence of kernels $K_i \subset H' \subset H \cong \mathbb{Z}_p$ with $\ell_i \to \infty$, so that $G_i = H/K_i$ has unbounded p-power order while displacements on a chart ball tend to zero. A finite cyclic group has no such nested open subgroups. Separately, any invariant map through a free $\mathbb{Z}/p$-orbit space has degree divisible by p (transfer), so there is no degree-one averaged pinch.

Conclusion: The proof does not incorrectly rule out finite cyclic actions. Break is at §8 geometric data, before Theorem 4.3. **Sanity PASS.**

Break-point matrix

For each model: first lemma/hypothesis that fails when one tries to copy the proof.
– means the step still makes sense; the chain stops at the first **BREAK**.

Model	Faithful $\mathbb{Z}_p$?	First BREAK	Later notes
Topological n-manifold (theorem setting)	Conjecturally no; paper claims no	*(none — target of proof)*	Full chain

Model	Faithful $\mathbb{Z}_p$?	First BREAK	Later notes
Finite $\mathbb{Z}/p$ on $\mathbb{R}^2$	N/A (finite)	Unbounded tail K_i; no deg-1 Haar pinch	Transfer degree
$\mathbb{Z}_p \times \mathbb{R}^n$ (left translation on $\mathbb{Z}_p$)	Yes	**Not a manifold** / no Euclidean chart (Lemma 8.1 chart)	Locally $\mathbb{Z}_p \times \mathbb{R}^n \not\simeq \mathbb{R}^{n+k}$
p-adic solenoid Σ_p	Yes (standard free action; orbit S^1)	**Not a manifold**; no chart ball in $\mathbb{R}^n$; no $S^n \times \mathbb{CP}^2$ from a coordinate compactification	Averaging to S^1 exists; Poincaré–Lefschetz dual blocks of Lemma 7.4 need a combinatorial manifold
Menger compactum μ^n (Dranishnikov 1989)	Yes — free action of any zero-dimensional compact group	**Not a manifold**; Newman fails; no chart; Lemma 7.4 triangulation fails	Orbit dim can equal n; irrelevant because chart already fails
Menger sponge / curve μ^1	Yes	Same as μ^n	Classic Cantor-like 1-dimensional universal curve
Raymond–Williams compactum	Yes; $\dim X = n$, $\dim X/\mathbb{Z}_p = n+2$	**Not a manifold**; chart / PL dual cells fail	Historically tied to Yang dim-raising; this paper does not use that obstruction
ENR homology n-manifold (Bryant–Ferry–Mio–Weinberger; Bing dogbone $\times\mathbb{R}$)	**Unknown** whether faithful $\mathbb{Z}_p$ can exist	**OPEN EDGE**	Local duality may partially survive; paper's Lemmas 7.3–7.5 and 12.1–12.2 use **combinatorial triangulations of manifolds**. Homology manifolds need not be triangulable in this sense

Detailed sponge notes

1. Dranishnikov Menger actions (including the sponge)

Fact: Every Menger manifold (in particular μ^n) admits a free continuous action of $\mathbb{Z}_p$ (Dranishnikov, *Math. USSR Izv.* 32 (1989)).

Why the Dabrowski argument stops:

1. μ^n is not locally Euclidean ⇒ Lemma 8.1's chart (image containing $B(0,4)$) is undefined.
2. Haar coordinate average F_0 is an average of points in $\mathbb{R}^n$, not an abstract barycenter on a compactum.
3. $W = S^n \times \mathbb{CP}^2$ is built from the **compactified coordinate sphere**. There is no such sphere for a Menger chart.
4. Lemma 7.4 needs a finite combinatorial triangulation of a compact oriented manifold with dual blocks. Menger compacta are not manifolds.
5. Newman's theorem (used in Corollary 1.2, not in Theorem 1.1's realization) is a manifold theorem.

The divisibility half (Theorem 4.3) is stated for an abstract compact free C_p-space T. That half **could** be copied on a Menger free action, producing p-divisible tails in $L_4(A_G(T))$. The contradiction never fires because realization of signature 1 never starts. **"Proves too much?"** No.

2. Solenoid

The inverse-limit circle $\Sigma_p = \varprojlim(S^1 \xrightarrow{\times p} S^1)$ carries a free $\mathbb{Z}_p$-action with orbit space S^1.

- Orbit space is a manifold; the **total space** is not.
- One can average maps to circles, but the paper's degree-one pinch is a compactified Euclidean chart map $s_\rho \circ F_0$.
- Break: chart + $S^n \times \mathbb{CP}^2$ + dual-block Poincaré–Lefschetz.

3. Raymond–Williams

These examples show dim-raising for $\mathbb{Z}_p$ actions on **compacta**. The Dabrowski proof **does not use** Yang's cohomological-dimension relation. On RW spaces the Euclidean PL pipeline never starts. They neither confirm nor refute the L-theory steps; they only show that wild orbit spaces exist — which the paper already allows, since it never bounds $\dim Y$.

4. Homology manifolds (edge)

Topological manifolds ⇒ ENR homology manifolds, but not conversely in high dimensions. Local Čech duality and orientation sheaves can exist without Euclidean charts.

Paper ingredient	Homology manifold?
Newman	Partial analogues exist; not automatic
Open $O \subset \mathbb{R}^n$ containing $B(0,4)$	**Fails** without resolution/chart
Haar coordinate average	Needs vector space structure
Combinatorial dual blocks (Lemma 7.4)	**Fails** without a triangulation of a manifold
Abstract $L_4(A_G(T))$ divisibility	Could make sense for a free C_p-space T
$\mathbb{CP}^2$ evaluation	Needs the product Poincaré manifold $S^n \times \mathbb{CP}^2$

If a free $\mathbb{Z}_p$ action on an ENR homology manifold were found, one would check whether some substitute for Lemmas 7–12 still produces a signature-one tail. That is a **stress region**, not a present counterexample to the written theorem.

Shared Euclidean ingredients (why sponges fail the same way as OpenAI)

Both proofs need, in some packaging:

1. Newman rigidity (corollary / reduction).
2. Local Euclidean charts.
3. An integral signature (Witt sheaf vs L_4 tail / $\mathbb{CP}^2$).
4. Arbitrarily small open subgroups of Homeo (Haar averaging / unbounded finite quotients).

Sponges break (1)–(2) immediately, so they do not test (3)–(4). That is the correct scope for a manifold theorem. See `05_vs_openai.md`.

Public responses (Q3)

Public responses to the Dabrowski Hilbert–Smith preprint (Q3)

AI disclaimer: These notes are AI-assisted and AI-generated using Cursor, with Grok, Anthropic, and OpenAI models, under the direction of Alan J. Coppola.

Search date: 2026-10-08
Paper: A. Dabrowski's AI agents, *Integral tail signatures and the Hilbert–Smith conjecture*, 2026-10-08, in adbrw/HS_proof.

Executive finding

As of the morning of 2026-10-08 there is **no published technical rebuttal or confirmation** of this proof's correctness. The repository is hours old (`created_at` 2026-10-07T08:08:15Z; paper commit 2026-10-08 09:06:46 UTC). Coverage, if any, is announcement-level. This Sherlock pass is therefore an independent first read, not a digest of consensus.

Lean status claimed by the author: a Lean 4 / Mathlib formalization of Corollary 1.2 (`HSFormal/`, 238 modules), with `Challenge.lean` / `Solution.lean` for comparator and SafeVerify. Claimed axioms: `propext`, `Classical.choice`, `Quot.sound`. **This audit did not rerun the checkers.**

Sources checked (2026-10-08)

Venue	Finding
GitHub adbrw/HS_proof	Public repo; 0 stars at vendor time; README states Thm 1.1 / Cor 1.2 and the Lean claims
MathOverflow	No hit on this preprint as of search (OpenAI Hilbert–Smith thread 515824 was closed as work-checking; unrelated to Dabrowski)
arXiv	No matching submission found under this title at search time; paper lives on GitHub as `HSproof.pdf`
Wikipedia / Tao 2011 blog / classical	Background on the conjecture only; no 2026 Dabrowski discussion
MO 32656	
Third-party HTML paste of the README	A garbled copy listed `HilbertSmithNegK.lean` and `HSproof.pdf` among "axioms." The **vendored** README does not. Treat that paste as unreliable; settle the axiom list by `#print axioms`

What the upstream README claims (not independently verified)

- Lean `v4.35.0-rc3`; mathlib `1a547d8…`; TauCeti `c7af81f`.
- `lake env lean` axiom print of `HSFormal.hilbertSmith`.
- Comparator last line: "Your solution is okay!"; SafeVerify: "SafeVerify check passed."
- Resource needs: Linux $\geq$ 5.19 with Landlock, ( )15 GB disk, 16 GB RAM.

Disclaimer in the PDF (p. 16)

The paper states that various LLMs (Anthropic, OpenAI, open-source) were used under human guidance at every stage; "the core idea and Lean4 formalization were obtained by Claude Opus 5.5 agents"; budget "below US$1,000." That is process context, not a correctness argument.

What to watch next

1. An independent `#print axioms HSFormal.hilbertSmith` from a clean `lake build`.
2. Expert notes from controlled topology / L-theory (Ranicki, Carlsson–Pedersen users) on §§2–6 and §§10–12.
3. Comparison with the OpenAI Witt-sheaf argument (same obstruction class; different packaging) — `05_vs_openai.md`.
4. Any MathOverflow / arXiv appearance under a non-announcement title.

Implication for this Sherlock session

Public silence on *technical* content means grades in `02_audit_ledger.md` are provisional expert-prep. Target C is `WEAK` primarily because the Lean certificate was not rerun here, not because a false axiom was found in the PDF.

Versus OpenAI

Dabrowski vs OpenAI: same obstruction class, different packaging

AI disclaimer: These notes are AI-assisted and AI-generated using Cursor, with Grok, Anthropic, and OpenAI models, under the direction of Alan J. Coppola.

Both preprints claim the Hilbert–Smith conjecture in every finite dimension by excluding faithful continuous $\mathbb{Z}_p$-actions on topological manifolds. Neither uses Yang's orbit-dimension raising, and neither is Pardon's 3-manifold mapping-class argument. They sit in the same **family**: integrality versus p-divisibility of a signature / Witt invariant manufactured from a Euclidean chart and a small open subgroup of Homeo.

OpenAI audit: `docs/Hilbert_Smith_Audit.pdf` (this repo, 2026-09-23 preprint).
Dabrowski audit: this directory.

One-line mechanisms

	OpenAI (2026-09-23)	Dabrowski (2026-10-08)
Invariant	Sheaf Witt group $E_d(S^d)$ of continuous polyhedral forms	Ultimate quadratic L_4 of an asymptotic category $A_G(T)$
Integrality	Fixed denominator lattice $L_d^{-1}\mathbb{Z}\,\overline{u}_d$, $L_d = 2^{e_d}$	Ordinary integer signatures; tail in $\mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})}$
p-input	p^k equal character pieces summing to $4\overline{u}_d$	Permutation form $\tau = \mathbb{Q}[G_i/P_i]$ locally p copies of id; $\nu^s = 0$
Contradiction	$4/p^k \notin L_d^{-1}\mathbb{Z}$ for $p^k > 4L_d$	$1 \in p\mathbb{Z}$
Test space	Odd-dimensional sphere S^d after stabilizing $O \subset \mathbb{R}^n$	$S^n \times \mathbb{CP}^2$; iterated boundaries recover $\mathbb{CP}^2$
Averaging	Haar-average a degree-one pinch through $Y = W^+/H$	Haar-average **coordinates** in a chart; pinch f on the compactified S^n
Formalization	Not found in the OpenAI dump	Claimed Lean 4 certificate (`HSFormal/`, 238 modules) — not rerun here

---

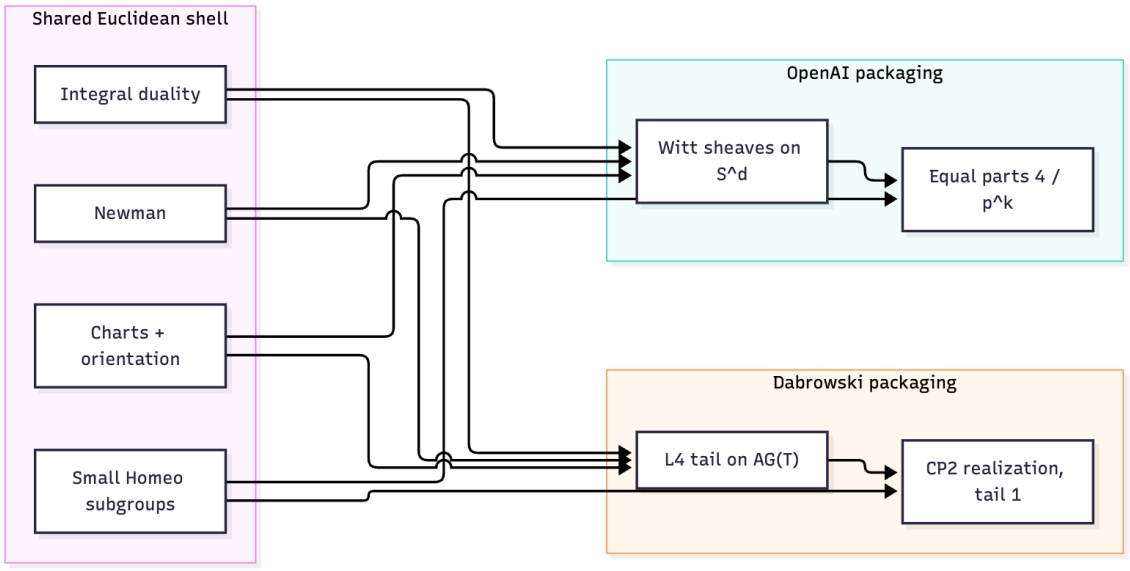
Shared Euclidean ingredients

These four properties are what sponges lack and what both proofs spend their geometric budget on:

1. **Newman rigidity.** Small compact Lie actions that barely move a chart ball are trivial. Used by OpenAI in chart reduction; by Dabrowski in Corollary 1.2 (after p-adic exclusion) via [Pa19].
2. **Local Euclidean charts + orientation.** OpenAI: invariant $O \subset \mathbb{R}^n$, stabilize into S^d. Dabrowski: free point, chart containing $B(0,4)$, compactify to S^n, product with $\mathbb{CP}^2$.
3. **Local duality / integral signature.** OpenAI: Verdier duality + Balmer triangular Witt + semialgebraic comparison. Dabrowski: controlled Poincaré–Lefschetz dual cells + Ranicki Poincaré pairs + Karoubi localization.

4. **Arbitrarily small open subgroups of** Homeo**.** Both need a tail of $\mathbb{Z}_p$ small enough that Haar averaging preserves degree one (OpenAI) or produces a signature-one $\mathbb{CP}^2$ germ (Dabrowski), and fine enough that finite quotients are unbounded p-powers.

On Menger / solenoid / Raymond–Williams spaces, **(1)–(2) fail first** for both arguments. That is why neither "proves too much."



Where they diverge

Topic	OpenAI	Dabrowski
Dimension shift	Odd stabilize $d > n$ so $\pi_*(S^d)$ is Serre-rationally concentrated	Keep dimension n; add $\mathbb{CP}^2$ to land in L_4 after n boundaries
Character theory	Finite partitions of p-power roots of unity on the punctured target	Equivariant signature characters of $\mathbb{Q}[G_i]$; cyclotomic Lemma 3.1
Lattice	Uniform 2-power denominator from $\mathcal{P} \to \mathcal{T}$ comparison	No denominator bookkeeping; integers from rational Witt / L_4
Orbit space	Explicit test $S^d \to Y \to S^d$	Invariant pinch on S^n; orbit space of T is a covering, not the manifold quotient
Cited black boxes	Balmer–Walter 2002; Woolf; Hardt	Carlsson–Pedersen 1995; Ranicki 1980/89/92/99; Balmer–Schlichting 2001; Weibel negative K
Weak spots (this repo's grades)	Dense doubling; BW localization; character coproducts; equal-parts isometry	Tail extraction / NegK; MV nilpotence; graph corner; $\mathbb{CP}^2$ evaluation; Lean axiom print

Topic	OpenAI	Dabrowski
Boundary	Statement is without boundary	Paper allows boundary; Lean has both forms

---

What a common research question looks like

If both arguments are pointing at the same rigidity of Euclidean space, the mathematical question is **not** "which PDF is correct?" but:

Which of (1)–(4) is the essential obstruction to continuous $\mathbb{Z}_p$-actions, and which can fail independently?

That is the MathOverflow rewrite in `docs/POST_DRAFTS.md`: a question about Euclidean structure, with both preprints as motivation.

Audit status comparison (targeted, 2026-10-08)

	OpenAI	Dabrowski
FAIL	0	0
Headline WEAK	4 lemmas	5 clusters (NegK/tail, MV nilpotence, graph corner, $\mathbb{CP}^2$ signs, Lean axioms)
Sponge over-claim	No	No
Finite $\mathbb{Z}/p$ sanity	Breaks at averaged-test	Breaks at unbounded tail / deg-1 Haar
Lean	None	Claimed; not independently printed here

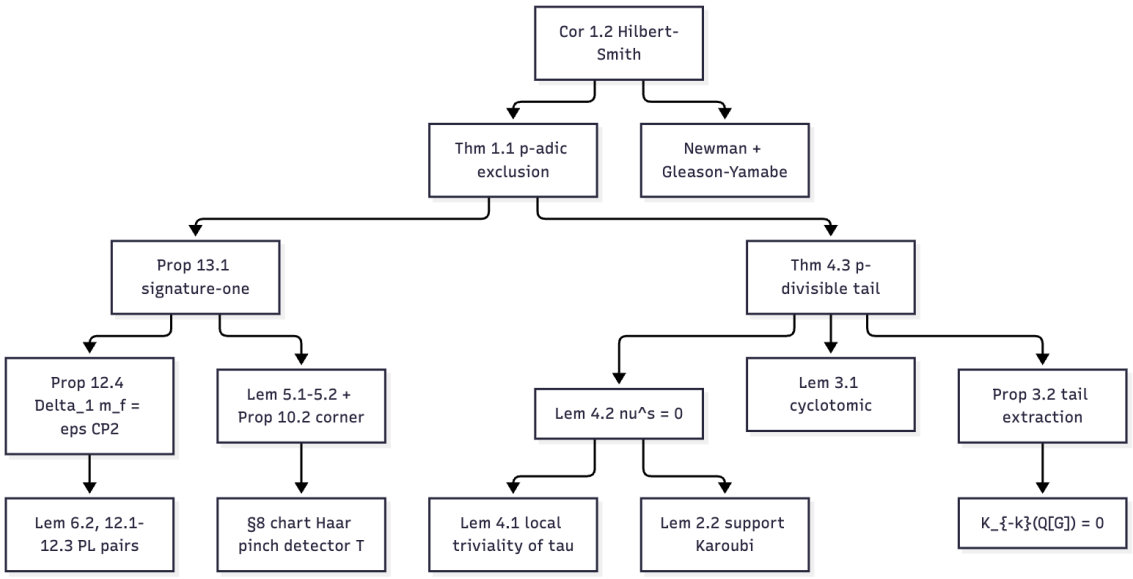
Lemma dependency ledger

Lemma dependency ledger (Dabrowski)

AI disclaimer: This ledger is AI-assisted and AI-generated using Cursor, with Grok, Anthropic, and OpenAI models, under the direction of Alan J. Coppola.

Extracted from the vendored PDF `HSproof.pdf` (17 pp.) and sparse Lean headers. There is no public LaTeX tree; labels below are PDF lemma numbers.

Core dependency chain (contradiction)



Targeted audit nodes (by Sherlock target)

Target	PDF objects	Sparse Lean
A	Lem 3.1, Prop 3.2, Thm 4.3, Prop 12.4, Prop 13.1	`AlgebraicReduction.lean`, `EquivariantSignature.lean`, `HilbertSmithNegK.lean`
B	Lem 2.2, 4.1, 4.2; §8 Haar pinch; Lem 5.1–5.2; Prop 10.2	`ChartData.lean`, `CharacterDetector.lean`, `CornerClass.lean`, `ControlledDuality.lean`
C	Challenge/Solution vs Thm 1.1 / Cor 1.2	`Challenge.lean`, `Solution.lean`, `Statement.lean`, `HilbertSmith*.lean`
Sanity	Unbounded K_i; deg-1 Haar	`ChartData.lean` comments on displacement $D < 1/64$

External citations used as black boxes

Citation	Role in proof
[CP95] Carlsson–Pedersen	Karoubi localization, connecting maps, pair description of boundaries
[Ran80I], [Ran80II]	Algebraic Poincaré pairs and unions
[Ran89], [Ran92]	Additive / lower L-theory; ultimate decoration; $L(Q) \cong L(\mathrm{Kar}\, Q)$
[Ran99]	Controlled Poincaré–Lefschetz, dual cells
[BS01] Balmer–Schlichting	Idempotent splitting in the bounded derived category of Karoubi completion
[Wei13]	$K_{-k}(\mathbb{Q}[G]) = 0$ for finite G
[Pa19] Pardon survey	Standard reduction; Uniform Newman
[Gle52], [Yam53]	No small subgroups $\Rightarrow$ Lie
[Rob75]	Pointwise finite families $\Rightarrow$ finite image (free point)
[Hat02]	Simplicial approximation; degree classifies maps $S^n \to S^n$ for $n \geq 1$

Lean engineering (not in the 17-page PDF)

Module cluster	Role
`Inputs/GY`, `Inputs/Lemma7`, `Newman/`	Gleason–Yamabe, Tao 2011 Lemma 7, Newman prime order
`LTheory/Model/` including `NegK/High/`	Concrete lower L-theory; `NegKHighAll`
`Brouwer/`	Five MIT-licensed files (upstream README)

Sherlock session log

Sherlock session — Dabrowski Hilbert–Smith preprint

AI disclaimer: This session log is AI-assisted and AI-generated using Cursor, with Grok, Anthropic, and OpenAI models, under the direction of Alan J. Coppola.

Date: 2026-10-08
Workspace: `/Users/ajjc/proj/Hilberts-5th-OAI`
Protocol: Sherlock Mode (6 steps) + plan `dabrowski_hs_sherlock`

Step 1 — candidate failure sources (wide net)

1. Newman / chart reduction (same family as OpenAI `prop:chart-reduction`).
2. Integral tail construction: L_4 / Witt / Karoubi / Ranicki Poincaré pairs.
3. p-divisibility / transfer / averaging on finite covers of T.
4. Negative-K / `NegKHighAll` / decoration-map bijectivity.
5. Lean vs PDF gap: `sorry`, extra axioms, statement mismatch.
6. Homotopy / duality signs (cylinder / dual-block foot-guns).
7. Overclaim on sponges/solenoids if hypotheses are weaker than manifolds.

Step 2 — distill (most likely)

- **A.** Signature-tail lattice / integer signatures independent of p-power.
- **B.** Transfer/averaging multiplies by p without leaving the integral setting.
- **C.** Lean certificate actually proves the published theorem.

Step 3 — investigate

- Vendored `HSproof.pdf` (SHA-256 `e305fd47…`) and sparse Lean from https://github.com/adbrw/HS_proof at `abb0c44`.
- Line-by-line on load-bearing PDF lemmas (not all 238 Lean modules).
- Same sponge matrix as OpenAI; comparison table in `05_vs_openai.md`.

Step 4 — analysis

- **Mechanism:** integer signature tail: realization gives $\sigma_{\text{tail}} = 1$; divisibility gives eventually $p \mid \sigma$; hence $1 \in p\mathbb{Z}$. Same obstruction class as OpenAI, packaged as Ranicki $L_4 + \mathbb{CP}^2$ rather than Witt sheaves on S^d.
- **Audit (A–C):** 0 FAIL on PDF load-bearing lemmas. WEAK: Prop. 3.2 / NegK, Lemma 4.2 MV nilpotence, graph corner §§10–11, Prop. 12.4 signs, Lean axiom print.
- **Sanity:** finite $\mathbb{Z}/p$ breaks at unbounded tail / degree-one Haar.
- **Sponges:** Menger/solenoid/RW break at chart / PL dual cells; no overclaim. Homology manifolds OPEN EDGE.
- **Public:** no technical rebuttal as of 2026-10-08 morning. Garbled third-party README paste listed `HSproof.pdf` as an axiom; vendored README does not.

Step 5 — follow-ups

1. Independent `#print axioms HSFormal.hilbertSmith` after `lake build`.

2. Expert check of Lemma 4.2 against [CP95] ultimate L Mayer–Vietoris.
3. Expert check of Prop. 12.4 iterated boundary evaluation.
4. MO rewrite as Euclidean-structure question citing both proofs (not work-checking).

Step 6 — cleanup

`/tmp/hs_dabrowski` clone may be deleted after confirming `dabrowski/` is complete. No temporary diagnostic code was added to the proof. Full `HSFormal/` (238 modules) and `HilbertSmithLean4.zip` were **not** copied into the repo on purpose (sparse vendor).

Deliverables

- `dabrowski/PROVENANCE.md`, `dabrowski/paper/`, `dabrowski/lean/` (sparse)
- `dabrowski/docs/01`–`05`, lemma ledger, this log
- `dabrowski/docs/Dabrowski_HS_Audit.pdf`
- Root `README.md` two-proof index; `docs/POST_DRAFTS.md` MO rewrite

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Directory README

Dabrowski Hilbert–Smith reverse verification

Sherlock-mode audit of A. Dabrowski's 2026-10-08 preprint *Integral tail signatures and the Hilbert–Smith conjecture*, with a sparse vendor of the claimed Lean 4 certificate.

Author: Alan J. Coppola
AI disclaimer: This audit is AI-assisted and AI-generated using Cursor, with Grok, Anthropic, and OpenAI models, under the author's direction. See `../docs/AI_DISCLAIMER.md`.

Upstream: https://github.com/adbrw/HS_proof
Vendored at: `abb0c44d58d711235f20b666ef422a4ea87b9add`
PDF SHA-256: `e305fd474909694af1e032ab863d26411f5e9fd9c38e7001f5fac14fe19e5e75`

Main deliverable: `docs/Dabrowski_HS_Audit.pdf`

Rebuild (needs `pandoc` + `tectonic`): `python3 scripts/build_dabrowski_audit_pdf.py`

This directory sits **alongside** the OpenAI audit (`paper/`, `docs/`). Comparison: `docs/05_vs_openai.md`.

Repository README

Hilbert–Smith reverse verification + homology-manifold boundary

Sherlock-mode audits of two 2026 preprints claiming the Hilbert–Smith conjecture in every finite dimension, plus a **relative** geometric-topology program that maps those arguments across homology / cohomology manifolds (HM-1..HM-6) and isolates an open construction target when $\dim_{\mathbb{Z}} X = n$ but $\dim X = \infty$.

Both 2026 arguments sit in the same family — **integrality versus p-divisibility of a signature / Witt invariant** after a Euclidean chart and Haar averaging — not Yang dim-raising and not Pardon's 3-manifold mapping-class theorem.

Author: Alan J. Coppola
AI disclaimer: This work is AI-assisted and AI-generated using Cursor, with Grok, Anthropic, and OpenAI models, under the author's direction. See `docs/AI_DISCLAIMER.md`.
Repo: https://github.com/ajjcoppola/hilberts-5th-oai

Homology manifolds (HM-1..HM-6)

Paper: Alan J. Coppola, *p-Adic Transfer, Signature Tails, and Homology Manifolds* (18 Oct 2026 draft) —
`papers/padic-homology-manifolds/main.pdf`

Class	What it is	Faithful continuous $\mathbb{Z}_p$?
HM-1	Non-*clc* / coefficient-mismatch Čech cohomology manifolds	Counterexample *mechanism* (outside 2026 scopes)
HM-2	Shrinking necklace / junction $\mathbb{Z}$-homology manifolds	**Explicit** counterexample (local homology fails)
HM-3	Connected polyhedral L-homology manifolds	**No** — reduces to manifold case (Thm C)
HM-4	Resolvable ENR / nonempty manifold stratum	**No** — chart or $X \times \mathbb{R}^2$ (Thm D)
HM-5	BFMW ENR, $X_{\text{man}} = \emptyset$, Quinn $I(X) \neq 1$	**No** (relative to OpenAI ENR rephrase or Dabrowski k-tower); bounded-dim inverse limits blocked
HM-6	Non-ENR: $\dim_{\mathbb{Z}} X = n < \infty$, $\dim X = \infty$	**Open candidate** — escapes E/F/G(2); Borel-join EW+Williams scaffolding; $clc_{\mathbb{Z}}$ glue still open

Honesty: Exclusion on manifolds/ENRs is **relative** to OpenAI/Dabrowski geometric engines. This repo does **not** claim a standalone classical Hilbert–Smith proof. HM-6 is a construction program, not a claimed existence theorem.

Homology-manifold deliverables

Deliverable	Path
Paper PDF / TeX	`papers/padic-homology-manifolds/`
Main slides	`presentations/padic-homology-2026/inde`
Invariants deck (Quinn / Witt / Dabrowski + geometric germ)	`presentations/invariants-quinn-witt-da`
Reviewer checklist / pack	`presentations/padic-homology-2026/`
Campaign + ledger + HM notes	`counterexamples/hilbert-smith/` (`HOMOLOGY_MANIFOLDS.md`, `CAMPAIGN.md`)
Model catalog (incl. E4 = HM-6)	`counterexamples/models/CATALOG.md`
CoppolaHS Lean	`lean/CoppolaHS/` (`CyclotomicTower`, `QuinnIndex`, `DranishnikovHM6`, …)
HM-6 harness / loop	`scripts/hm6_equivariant_ew_harness.py`, `HM6_LOOP_LOG.md`
Finish-line plan (Trunks A/B + glue P3)	`plans/HM6_FINISH_LINE.md`
Literature library (arXiv PDFs + construction notes)	`library/README.md`
Outreach / collaborators	`docs/OUTREACH_COLLABORATORS.md`
Post drafts (MO / X / LinkedIn)	`docs/POST_DRAFTS.md`

Checkpoints (git tags)

Tag	Contents
`checkpoint/hm6-2026-10-18`	Archive paper, CoppolaHS Lean, HM-6 harness/loop, counterexample campaign
`checkpoint/invariants-deck-2026-10-19`	Quinn / Witt / Dabrowski invariants slides (KaTeX + geometric germ narrative)

Notes: `docs/CHECKPOINT_HM6_2026-10-18.md`.

HM-6 construction sketch

Open frontier where OpenAI ENR rephrase, Dabrowski k-tower, and bounded-dimension Euler obstruction all fail:

- **Trunk A:** growing lens / solenoid / circle-bundle stages (`library/constructions/trunk_A_lens_solenoid`
- **Trunk B:** finite recursive C_{p^i} + Edwards–Walsh Borel-join (`library/constructions/trunk_B_cyclic_EW.`
- **Shared finish P3:** $clc_{\mathbb{Z}}$ Poincaré glue (`library/constructions/glue_P3_clc.md`)

Finite-stage algebraic gates G1–G4 **PASS** symbolically; continuous glue remains **EDGE / open**.

2026 preprint audits

Proof	Date	Packaging	Audit PDF
OpenAI	2026-09-23	Witt/signature lattice on S^d; equal parts $4/p^k$	`docs/Hilbert_Smith_Audit.pdf`
Dabrowski	2026-10-08	Integral tail in $L_4(A_G(T))$; $\mathbb{CP}^2$ realization vs $p\mathbb{Z}$	`dabrowski/docs/Dabrowski_HS_Audit.pdf`

Side-by-side: `dabrowski/docs/05_vs_openai.md`.
Geometric packaging of the invariants: `presentations/invariants-quinn-witt-dabrowski/`.

Counterexample protocol: `counterexamples/README.md`.

OpenAI (2026-09-23)

Upstream: https://github.com/openai/math (vendored at `adc7f1241b42e322a6451854ab7e4b4c146bf78a`)

Quick answers

Q	Answer
1. Near-id exclusion	Small open $\mathbb{Z}_p$ subgroups + Haar averaging give a degree-1 test through the orbit space; finite groups cannot.
2. Geometric vs algebraic	Mostly Witt/signature lattice arithmetic; thin geometric shell (Newman, chart, Serre).
3. Public responses	Process news only (2026-10-07); no technical rebuttal; no Lean for this proof.
4. Structure of $\mathbb{R}^n$	Signature-rigid under continuous $\mathbb{Z}_p$ orbit factorization after odd stabilization into S^d.
5. Menger sponges	Theorem false for sponges; proof breaks at chart/Newman — correctly scoped, does not overclaim.
6. Orbit factoring?	Necessary but not the whole story; contradiction is equal parts vs fixed denominator lattice.

Targeted audit: no `FAIL`; several `WEAK` (dense doubling, Balmer–Walter bridge, character coproducts, equal-parts isometry).

- `paper/` — vendored LaTeX + PDF + provenance
- `docs/01_proof_overview.md` — Q1, Q2, Q4, Q6
- `docs/02_audit_ledger.md` — lemma grades
- `docs/03_sponge_stress_test.md` — Q5 + sanity
- `docs/04_public_responses.md` — Q3
- `docs/lemma_dependency_ledger.md`
- Rebuild: `python3 scripts/build_audit_pdf.py`

Dabrowski (2026-10-08)

Upstream: https://github.com/adbrw/HS_proof (vendored at `abb0c44d58d711235f20b666ef422a4ea87b9add`)

Quick answers

Q	Answer
1. Near-id exclusion	Small open tails of $\mathbb{Z}_p$ Haar-average a degree-1 pinch in a chart; unbounded finite quotients feed an L_4 tail of signature 1, which divisibility forces into $p\mathbb{Z}$.
2. Geometric vs algebraic	Mostly ultimate quadratic L-theory (Karoubi / Ranicki); thin Euclidean shell (chart, PL dual cells, $\mathbb{CP}^2$).
3. Public responses	None technical as of 2026-10-08 morning; claimed Lean 4 certificate not rerun here.
4. Structure of $\mathbb{R}^n$	Signature-tail rigid: a Euclidean germ realizes $\mathbb{CP}^2$ with $\sigma = 1$, which cannot be eventually p-divisible.
5. Menger sponges	Breaks at chart / combinatorial dual cells — correctly scoped.
6. Orbit factoring?	Necessary but not sufficient; contradiction is $\sigma_{\text{tail}} = 1$ vs eventual p-divisibility.

Targeted audit: no `FAIL` on load-bearing PDF lemmas; `WEAK` on NegK/tail extraction, Mayer–Vietoris nilpotence, the graph corner, $\mathbb{CP}^2$ evaluation, and Lean `#print axioms`.

Local `#print axioms HSFormal.hilbertSmith` (optional palomar clone): `propext`, `Classical.choice`, `Quot.sound` — see `dabrowski/PROVENANCE.md`, `dabrowski/lean-extension/BUILD_STATUS.md`.

- `dabrowski/paper/` — vendored `HSproof.pdf` + Challenge/Solution + lake files
- `dabrowski/lean/` — sparse load-bearing modules (not the full 238)
- `dabrowski/docs/01_proof_overview.md`
- `dabrowski/docs/02_audit_ledger.md`
- `dabrowski/docs/03_sponge_stress_test.md`
- `dabrowski/docs/04_public_responses.md`
- `dabrowski/docs/05_vs_openai.md`
- Rebuild: `python3 scripts/build_dabrowski_audit_pdf.py`

MathOverflow / posts

Paste-ready Euclidean-structure rewrite and HM-6 research question (not work-checking): `docs/POST_DRAFTS.md`.